Written Section Technology Value Map

Facts:

- The self-service user portal is difficult to use for most students (this is based on my personal experience as a more technically-advanced user, as well as several students complaining about the portal in person and online forums).
- The campus has just recently reported they are opening a brand new physical plant location that is much larger.

Assumptions:

- The new updates/additions to the expanding physical plant facility should help ease many issues stated by students and faculty alike.
- The campus plants KNOWS this process moves slowly for both users and technicians yet has not done the work to fix these changes.

Open-ended Questions:

- How much would it cost to do a complete UX overhaul, and if the cost isn't absorbent, why hasn't it already been completed?
- What does the portal look like on the technicians/physical plant side? If the data isn't primarily hosted by us who actually does host this information and what is our client-vendor relationship like?
- What new challenges will be faced with a new physical plant location?

Reasoning

Identifying users, customers, support teams, etc.

Our users should be the physical plant team/staff, our customers will be students/campus faculty, our decision makers are the plant managers and budget/accounting managers, our support team is the physical plant staff/technicians and possible third party project staff/construction workers, and our vendors will be any companies who's physical or digital tools we use.

What the pain points are and why they were chosen.

- For step 1, a pain point to be noted is Experience. Work order submitted by email or phone is inconsistent and gives the requester no confirmation, so frustration is felt by the person reporting.

- In step 2, a possible pain point is Reliability. Manual data entry into the CMMS delays the start of everything which delays each following process.
- For step 3, a possible pain point is Speed, although automated system should have no real bottleneck, plant support / techs actually prioritizing the work orders can lead to delays.
- Step 4 works in tandem with step 3 (Speed), if there is assigning and scheduling without a clear view of the technicians workload, it can make dispatching a tech take much longer than needed
- For step 5, Speed / Visibility is another possible pain point, needing to rely on a mobile system to then dispatch the tech can have possible delays, additionally the system should update the requester as to who the tech is and when to expect arrival.
- For step 6 we have Cost / Visibility as possible pain point, for example instead of an automated system, using a spreadsheet inventory that shows empty inventory list late, forces the plant to then rush orders at premium prices.
- For step 7, it would seem that Risk the biggest possible pain point due to on-site work, with no additional safety systems in place that means safety and compliance steps aren't being tracked.
- For step 8, a possible pain point could be Reliability, while closing out a ticket should be a defined step that the CMMS already handles, systems are bound to break and there needs to be a backup solution.
- For step 9 we have Reliability as a possible pain point, while an automated system may be in place, systems can always break, and there needs to be another way ticket closings are sent out, otherwise the requesters often never learn the job is done.
- Finally for step 10 we have Visibility, with outdated or difficult to use inventory and payment systems in place, the delayed cost and asset updates can obscure true spend and asset condition.

Three Improvement Opportunities

- Modernizing the portal interface is our first priority. The self-service portal already captures requests and pushes automated updates, so value is being lost only at the surface. A UX redesign that has cleaner forms, mobile-friendly layout, fewer clicks would help lift adoption and cuts abandoned or misfiled requests without touching the working backend. It's high impact for relatively low cost because you're fixing basically the image, not rebuilding the system.
- Speeding up technician dispatch is next. This targets the speed pain at step 5, the biggest remaining delay. Mobile work orders exist, but the gap between a ready work order and a technician actually on site is what requesters feel most. Priority-based assignment, zone routing, and real-time technician availability shorten response time directly. This matters because dispatch speed is the metric the campus experiences as "how fast does the plant fix things."

- Integrate parts and inventory with the CMMS. This targets the cost pain at step 6. Inventory is still a spreadsheet, so none-in-stock alerts only appear after a tech arrives, forcing rush orders and repeat trips. Linking inventory to the CMMS reserves parts when the work order is created, cutting expedited-shipping premiums and second visits.

Executive Statement

By modernizing the request portal and accelerating technician dispatch, we can shorten repair response times and reduce rush-part costs — turning systems we already own into faster, more reliable service for campus without adding headcount.

- The goal is for the plant manager and accounting / budgeting manager to see this as a proposal to get more value from our current infrastructure rather than asking for new spend.